

It can be very difficult to perform statistical analysis over anonymous and/or self-reported data. This style of data reporting and collection has very few guardrails so it's very easy to have a variety of formats and styles used when inputting data. For example, the Grad Café survey uses a lot of drop-down boxes, which helps with formatting, but it also uses a lot of form fields. This allows too much variety to be reliably used without extensive data preprocessing. This can be highlighted by the GRE scores. While there are form fields for GRE, many times the GRE scores were placed in the notes instead. Also, even though the undergrad GPA field says it should be out of 4.0, there were many GPAs that were larger than 4.0. While that's technically possible, it is unlikely that very many people would have GPAs as high as 4.8, which we saw in the data. Between all these issues we end up dealing with unstructured data from the notes section, potentially inaccurate data from either typos or personal bias, and a variety of formats. This causes issues not just with data cleaning and preprocessing, but with the data analysis because too many incorrect or strangely formatted fields will cause a variety of issues with the analysis.

Another issue that comes from anonymous and/or self-reported data is the bias of the people submitting the data. Typically, the people that self-report to surveys have stronger feelings than the people who don't. This can cause some strange effects on the data, particularly for skewing and kurtosis. Additionally, the lack of data verification allows people to troll the page. A good example of this is entry 996532, which says the submitter was accepted to a Political Science PhD program at "Hot Piss University". The comment text then talks about urinals. Right before that, in entry 996531, is basically the same data but the school is "Brown Rice University" and the comment talks about a stir fry. Even these are mild compared to the very aggressive data in entry 995461. These are clearly not real schools or data, but they were pulled into the dataset by the web scraper and are hard to filter out without manually going through the data. All of these inconsistencies affect the analyses and conclusions for the dataset.